

Review Questions

The answers to the chapter review questions can be found in the Appendix.

1. Which of the following data types can be used in a `switch` expression? (Choose all that apply.)
 - A. `enum`
 - B. `int`
 - C. `Byte`
 - D. `long`
 - E. `String`
 - F. `char`
 - G. `var`
 - H. `double`
2. What is the output of the following code snippet? (Choose all that apply.)

```
3: int temperature = 4;
4: long humidity = -temperature + temperature * 3;
5: if (temperature >= 4)
6: if (humidity < 6) System.out.println("Too Low");
7: else System.out.println("Just Right");
8: else System.out.println("Too High");
```

 - A. Too Low
 - B. Just Right
 - C. Too High
 - D. A `NullPointerException` is thrown at runtime.
 - E. The code will not compile because of line 7.
 - F. The code will not compile because of line 8.
3. Which of the following data types are permitted on the right side of a `for-each` expression? (Choose all that apply.)
 - A. `Double[][]`
 - B. `Object`
 - C. `Map`
 - D. `List`
 - E. `String`
 - F. `char[]`
 - G. `Exception`
 - H. `Set`

4. What is the output of calling `printReptile(6)`?

```
void printReptile(int category) {  
    var type = switch(category) {  
        case 1,2 -> "Snake";  
        case 3,4 -> "Lizard";  
        case 5,6 -> "Turtle";  
        case 7,8 -> "Alligator";  
    };  
    System.out.print(type);  
}
```

- A. Snake
 - B. Lizard
 - C. Turtle
 - D. Alligator
 - E. TurtleAlligator
 - F. None of the above
5. What is the output of the following code snippet?

```
List<Integer> myFavoriteNumbers = new ArrayList<>();  
myFavoriteNumbers.add(10);  
myFavoriteNumbers.add(14);  
for (var a : myFavoriteNumbers) {  
    System.out.print(a + ", ");  
    break;  
}  
  
for (int b : myFavoriteNumbers) {  
    continue;  
    System.out.print(b + ", ");  
}  
  
for (Object c : myFavoriteNumbers)  
    System.out.print(c + ", ");
```

- A. It compiles and runs without issue but does not produce any output.
- B. 10, 14,
- C. 10, 10, 14,
- D. 10, 10, 14, 10, 14,
- E. Exactly one line of code does not compile.
- F. Exactly two lines of code do not compile.
- G. Three or more lines of code do not compile.
- H. The code contains an infinite loop and does not terminate.

6. Which statements about decision structures are true? (Choose all that apply.)
- A. A for-each loop can be executed on any Collections Framework object.
 - B. The body of a while loop is guaranteed to be executed at least once.
 - C. The conditional expression of a for loop is evaluated before the first execution of the loop body.
 - D. A switch expression that takes a String and assigns the result to a variable requires a default branch.
 - E. The body of a do/while loop is guaranteed to be executed at least once.
 - F. An if statement can have multiple corresponding else statements.
7. Assuming weather is a well-formed nonempty array, which code snippet, when inserted independently into the blank in the following code, prints all of the elements of weather? (Choose all that apply.)

```
private void print(int[] weather) {
    for(_____) {
        System.out.println(weather[i]);
    }
}
```

- A. `int i=weather.length; i>0; i--`
 - B. `int i=0; i<=weather.length-1; ++i`
 - C. `var w : weather`
 - D. `int i=weather.length-1; i>=0; i--`
 - E. `int i=0, int j=3; i<weather.length; ++i`
 - F. `int i=0; ++i<10 && i<weather.length;`
 - G. None of the above
8. What is the output of calling `printType(11)`?
- ```
31: void printType(Object o) {
32: if(o instanceof Integer bat) {
33: System.out.print("int");
34: } else if(o instanceof Integer bat && bat < 10) {
35: System.out.print("small int");
36: } else if(o instanceof Long bat || bat <= 20) {
37: System.out.print("long");
38: } default {
39: System.out.print("unknown");
40: }
41: }
```

- A. int
  - B. small int
  - C. long
  - D. unknown
  - E. Nothing is printed.
  - F. The code contains one line that does not compile.
  - G. The code contains two lines that do not compile.
  - H. None of the above
9. Which statements, when inserted independently into the following blank, will cause the code to print 2 at runtime? (Choose all that apply.)

```
int count = 0;
BUNNY: for(int row = 1; row <=3; row++)
 RABBIT: for(int col = 0; col <3 ; col++) {
 if((col + row) % 2 == 0)
 _____;
 count++;
 }
System.out.println(count);
```

- A. break BUNNY
  - B. break RABBIT
  - C. continue BUNNY
  - D. continue RABBIT
  - E. break
  - F. continue
  - G. None of the above, as the code contains a compiler error.
10. Given the following method, how many lines contain compilation errors? (Choose all that apply.)

```
10: private DayOfWeek getWeekDay(int day, final int thursday) {
11: int otherDay = day;
12: int Sunday = 0;
13: switch(otherDay) {
14: default:
15: case 1: continue;
16: case thursday: return DayOfWeek.THURSDAY;
17: case 2,10: break;
```

```
18: case Sunday: return DayOfWeek.SUNDAY;
19: case DayOfWeek.MONDAY: return DayOfWeek.MONDAY;
20: }
21: return DayOfWeek.FRIDAY;
22: }
```

- A. None, the code compiles without issue.
  - B. 1
  - C. 2
  - D. 3
  - E. 4**
  - F. 5
  - G. 6
  - H. The code compiles but may produce an error at runtime.
11. What is the output of calling `printLocation(Animal.MAMMAL)`?

```
10: class Zoo {
11: enum Animal {BIRD, FISH, MAMMAL}
12: void printLocation(Animal a) {
13: long type = switch(a) {
14: case BIRD -> 1;
15: case FISH -> 2;
16: case MAMMAL -> 3;
17: default -> 4;
18: };
19: System.out.print(type);
20: } }
```

- A. 3**
  - B. 4
  - C. 34
  - D. The code does not compile because of line 13.
  - E. The code does not compile because of line 17.
  - F. None of the above
12. What is the result of the following code snippet?
- ```
3: int sing = 8, squawk = 2, notes = 0;
4: while(sing > squawk) {
5:     sing--;
6:     squawk += 2;
```

```
7:    notes += sing + squawk;
8: }
9: System.out.println(notes);
```

- A. 11
- B. 13
- C. 23
- D. 33
- E. 50
- F. The code will not compile because of line 7.

13. What is the output of the following code snippet?

```
2: boolean keepGoing = true;
3: int result = 15, meters = 10;
4: do {
5:     meters--;
6:     if(meters==8) keepGoing = false;
7:     result -= 2;
8: } while keepGoing;
9: System.out.println(result);
```

- A. 7
- B. 9
- C. 10
- D. 11
- E. 15
- F. The code will not compile because of line 6.
- G. The code does not compile for a different reason.

14. Which statements about the following code snippet are correct? (Choose all that apply.)

```
for(var penguin : new int[2])
    System.out.println(penguin);
var ostrich = new Character[3];
for(var emu : ostrich)
    System.out.println(emu);
List<Integer> parrots = new ArrayList<Integer>();
for(var macaw : parrots)
    System.out.println(macaw);
```

- A. The data type of penguin is Integer.
- B. The data type of penguin is int.
- C. The data type of emu is undefined.
- D. The data type of emu is Character.
- E. The data type of macaw is List.
- F. The data type of macaw is Integer.
- G. None of the above, as the code does not compile.

15. What is the result of the following code snippet?

```
final char a = 'A', e = 'E';
char grade = 'B';
switch (grade) {
    default:
    case a:
    case 'B': 'C': System.out.print("great ");
    case 'D': System.out.print("good "); break;
    case e:
    case 'F': System.out.print("not good ");
}
```

- A. great
- B. great good
- C. good
- D. not good
- E. The code does not compile because the data type of one or more case statements does not match the data type of the switch variable.
- F. None of the above

16. Given the following array, which code snippets print the elements in reverse order from how they are declared? (Choose all that apply.)

```
char[] wolf = {'W', 'e', 'b', 'b', 'y'};
```

A.

```
int q = wolf.length;
for( ; ; ) {
    System.out.print(wolf[--q]);
    if(q==0) break;
}
```

B.

```
for(int m=wolf.length-1; m>=0; --m)
    System.out.print(wolf[m]);
```

C.

```
for(int z=0; z<wolf.length; z++)
    System.out.print(wolf[wolf.length-z]);
```

D.

```
int x = wolf.length-1;
for(int j=0; x>=0 && j==0; x--)
    System.out.print(wolf[x]);
```

E.

```
final int r = wolf.length;
for(int w = r-1; r>-1; w = r-1)
    System.out.print(wolf[w]);
```

F.

```
for(int i=wolf.length; i>0; --i)
    System.out.print(wolf[i]);
```

G. None of the above

- 17.** What distinct numbers are printed when the following method is executed? (Choose all that apply.)

```
private void countAttendees() {
    int participants = 4, animals = 2, performers = -1;
    while((participants = participants+1) < 10) {}
    do {} while (animals++ <= 1);
    for( ; performers<2; performers+=2) {}

    System.out.println(participants);
    System.out.println(animals);
    System.out.println(performers);
}
```

A. 6**B.** 3**C.** 4**D.** 5**E.** 10**F.** 9**G.** The code does not compile.**H.** None of the above

18. Which statements about pattern matching and flow scoping are correct? (Choose all that apply.)
- A. Pattern matching with an `if` statement is implemented using the `instance` operator.
 - B. Pattern matching with an `if` statement is implemented using the `instanceon` operator.
 - C. Pattern matching with an `if` statement is implemented using the `instanceof` operator.
 - D. The pattern variable cannot be accessed after the `if` statement in which it is declared.
 - E. Flow scoping means a pattern variable is only accessible if the compiler can discern its type.
 - F. Pattern matching can be used to declare a variable with an `else` statement.

19. What is the output of the following code snippet?

```
2: double iguana = 0;
3: do {
4:     int snake = 1;
5:     System.out.print(snake++ + " ");
6:     iguana--;
7: } while (snake <= 5);
8: System.out.println(iguana);
```

- A. 1 2 3 4 -4.0
 - B. 1 2 3 4 -5.0
 - C. 1 2 3 4 5 -4.0
 - D. 0 1 2 3 4 5 -5.0
 - E. The code does not compile.
 - F. The code compiles but produces an infinite loop at runtime.
 - G. None of the above
20. Which statements, when inserted into the following blanks, allow the code to compile and run without entering an infinite loop? (Choose all that apply.)

```
4: int height = 1;
5: L1: while(height++ <10) {
6:     long humidity = 12;
7:     L2: do {
8:         if(humidity-- % 12 == 0) _____;
9:         int temperature = 30;
10:        L3: for( ; ; ) {
11:            temperature++;
12:            if(temperature>50) _____;
13:        }
14:    } while (humidity > 4);
15: }
```

- A. `break L2 on line 8; continue L2 on line 12`
- B. `continue on line 8; continue on line 12`
- C. `break L3 on line 8; break L1 on line 12`
- D. `continue L2 on line 8; continue L3 on line 12`
- E. `continue L2 on line 8; continue L2 on line 12`
- F. None of the above, as the code contains a compiler error

21. A minimum of how many lines need to be corrected before the following method will compile?

```
21: void findZookeeper(Long id) {  
22:     System.out.print(switch(id) {  
23:         case 10 -> {"Jane"}  
24:         case 20 -> {yield "Lisa"}};  
25:         case 30 -> "Kelly";  
26:         case 30 -> "Sarah";  
27:         default -> "Unassigned";  
28:     });  
29: }
```

- A. Zero
- B. One
- C. Two
- D. Three
- E. Four
- F. Five

22. What is the output of the following code snippet? (Choose all that apply.)

```
2: var tailFeathers = 3;  
3: final var one = 1;  
4: switch (tailFeathers) {  
5:     case one: System.out.print(3 + " ");  
6:     default: case 3: System.out.print(5 + " ");  
7: }  
8: while (tailFeathers > 1) {  
9:     System.out.print(--tailFeathers + " "); }
```

- A. 3
- B. 5 1
- C. 5 2
- D. 3 5 1
- E. 5 2 1
- F. The code will not compile because of lines 3–5.
- G. The code will not compile because of line 6.

23. What is the output of the following code snippet?

```
15: int penguin = 50, turtle = 75;
16: boolean older = penguin >= turtle;
17: if (older = true) System.out.println("Success");
18: else System.out.println("Failure");
19: else if(penguin != 50) System.out.println("Other");
```

- A. Success
- B. Failure
- C. Other
- D. The code will not compile because of line 17.
- E. The code compiles but throws an exception at runtime.
- F. None of the above

24. Which of the following are possible data types for `friends` that would allow the code to compile? (Choose all that apply.)

```
for(var friend in friends) {
    System.out.println(friend);
}
```

- A. Set
- B. Map
- C. String
- D. `int[]`
- E. Collection
- F. `StringBuilder`
- G. None of the above

25. What is the output of the following code snippet?

```
6: String instrument = "violin";
7: final String CELLO = "cello";
8: String viola = "viola";
9: int p = -1;
10: switch(instrument) {
11:     case "bass" : break;
12:     case CELLO : p++;
13:     default: p++;
14:     case "VIOLIN": p++;
15:     case "viola" : ++p; break;
16: }
17: System.out.print(p);
```

- A. -1
- B. 0
- C. 1
- D. 2**
- E. 3
- F. The code does not compile.

26. What is the output of the following code snippet? (Choose all that apply.)

```
9: int w = 0, r = 1;
10: String name = "";
11: while(w < 2) {
12:     name += "A";
13:     do {
14:         name += "B";
15:         if(name.length()>0) name += "C";
16:         else break;
17:     } while (r <=1);
18:     r++; w++; }
19: System.out.println(name);
```

- A. ABC
- B. ABCABC
- C. ABCABCABC
- D. Line 15 contains a compilation error.
- E. Line 18 contains a compilation error.
- F. The code compiles but never terminates at runtime.**
- G. The code compiles but throws a `NullPointerException` at runtime.

27. What is printed by the following code snippet?

```
23: byte amphibian = 1;
24: String name = "Frog";
25: String color = switch(amphibian) {
26:     case 1 -> { yield "Red"; }
27:     case 2 -> { if(name.equals("Frog")) yield "Green"; }
28:     case 3 -> { yield "Purple"; }
29:     default -> throw new RuntimeException();
30: };
31: System.out.print(color);
```

- A. Red
- B. Green
- C. Purple
- D. RedPurple
- E. An exception is thrown at runtime.
- F. The code does not compile.

28. What is the output of calling `getFish("goldie")`?

```
40: void getFish(Object fish) {  
41:     if (!(fish instanceof String guppy))  
42:         System.out.print("Eat!");  
43:     else if (!(fish instanceof String guppy)) {  
44:         throw new RuntimeException();  
45:     }  
46:     System.out.print("Swim!");  
47: }
```

- A. Eat!
- B. Swim!
- C. Eat! followed by an exception.
- D. Eat!Swim!
- E. An exception is printed.
- F. None of the above

29. What is the result of the following code?

```
1: public class PrintIntegers {  
2:     public static void main(String[] args) {  
3:         int y = -2;  
4:         do System.out.print(++y + " ");  
5:         while(y <= 5);  
6:     } }
```

- A. -2 -1 0 1 2 3 4 5
- B. -2 -1 0 1 2 3 4
- C. -1 0 1 2 3 4 5 6
- D. -1 0 1 2 3 4 5
- E. The code will not compile because of line 5.
- F. The code contains an infinite loop and does not terminate.